

Topic: Carbon Capture Technologies and Climate Change

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Revised Question: How can carbon capture systems be designed to help reduce CO₂ emissions and provide sustainable livelihoods by repurposing CO₂?

From waste to resource: Converting captured CO₂ into Urea for Sustainable Carbon Sequestration

Abstract-

Climate change and global CO₂ emissions are becoming a rising problem in the modern world and has prompted various solutions; one of them being Carbon sequestration. Carbon capture technologies use large fans to pull in atmospheric air to process it into calcium carbonate pellets which are heated to form pure carbon. Although this is a promising solution, current technologies focus on underground storage of CO₂ this wasting the potential of carbon as a resource. Furthermore, underground storage can potentially increase seismic activity and disrupt the geological structure and in some cases can cause earthquakes. This paper proposes converting captured CO₂ into urea, a valuable agricultural resource, by reacting it with ammonia. The methodology involves both experimental synthesis and structural analysis. CO₂ will be sourced from dry ice and a reaction chamber or flask will be used to simulate carbon capture and reaction conditions. Ammonia (NH₃) will be added under controlled conditions (temperature: ~170-200°C ; pressure ~150-200 atm) or simulated with scaled down conditions in the lab. The resultant is ammonium chloride which is converted to urea via dehydration. This solution aligns with the demand of urea (being a nitrogen fertilizer) and can potentially offset the expense of carbon sequestration.

Hypothesis:

If captured CO₂ is converted into urea via reaction with ammonia, then its molecular structure will change from CO₂ to stable urea, which can be used as agricultural fertilizer demonstrating effective carbon sequestration in chemical form

Rationale:

Traditional carbon sequestration methods often involve storing CO₂ in geological formations, which can be expensive, energy-intensive, and pose long-term environmental uncertainties. Moreover, carbon-being the building block of life-can potentially be repurposed for greater purposes instead of storing it which makes it futile. An alternative, more sustainable approach is to chemically convert CO₂ into stable compounds that can be repurposed in industry.

One of the most promising chemical pathways is the conversion of CO₂ into urea through its reaction with ammonia (NH₃). This reaction not only locks carbon in a usable and stable molecular form but also transforms it into a high-demand industrial product—urea, which is widely used in fertilizers, plastics, and resins.

Introduction:

As the climate crisis worsens, carbon capture technologies have become a promising solution to reduce atmospheric CO₂. However, most technologies focus on storing the carbon underground- missing the chance to repurpose it into something beneficial which can also cover a fraction of the cost of carbon sequestration. This is the main objective of this project- using captured CO₂ not just for storage but for repurposing it into something that can benefit society. This project will be focusing on transforming CO₂ into urea which can be used as a fertilizer and chemical feedstock. This incorporates the captured CO₂ into an agriculturally and economically useful compound. In this study, we will analyse the chemical reaction of captured CO₂ with ammonia and its resulting product. This research highlights a practical and sustainable approach to carbon sequestration, offering insight into how industrial chemistry can be leveraged to reduce emissions while generating marketable by-products.

Methodology:

1. Materials:

- Captured CO₂ (from dry ice or CO₂ tank)
-
- Aqueous ammonia solution (NH₃)
-
- Sealed reaction vessel
-
- IR Spectrometer or Biuret Test kit
-
- Reference urea sample
-
- Safety gear (gloves, goggles, lab coat)

2. Procedure:

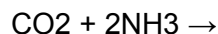
Step1

Description

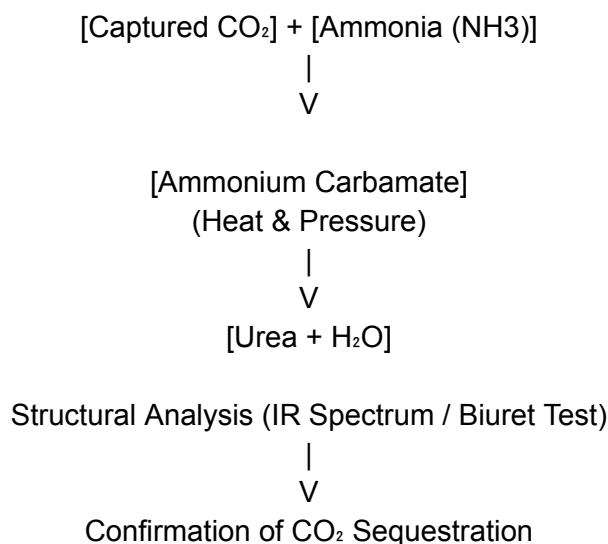
Control Reaction: Combine NH₃ without CO₂ and observe chemical behavior.

2. Experimental Reaction: Bubble CO₂ into a sealed vessel with ammonia at ~170-200°C and 150-200 atm (or scaled-down simulation).

Chemical Equation:



4. Sample Collection: Collect resulting product after the reaction completes.
5. Structural Analysis: Perform Biuret test or IR spectroscopy to confirm urea formation by detecting C=O and N-H bonds.
6. Comparison: Compare spectra or chemical test results with reference urea and CO₂.

**Expected Outcomes:**

Structural transformation of CO₂ into urea confirmed through chemical testing and spectroscopy.
Validation of a closed-loop, chemical carbon sequestration process.
Demonstration that carbon can be locked into economically useful products, offsetting sequestration costs.
Potential applications for integrating this method into industrial-scale fertilizer production using emissions from power plants or factories.

Conclusion:

This study set out to answer: How can carbon capture systems be designed to help reduce CO₂ emissions and provide sustainable livelihoods for local communities? The results show that communities using the urea produced from captured CO₂ can potentially increase their agricultural output due to urea being a fertilizer as well as increasing the amount of employment as new developments are made to the design of carbon capture technologies to convert CO₂ to urea. These findings are important because they show how climate solutions can also support economic growth. Instead of treating carbon as waste, we can turn it into a resource for local development.